Programme: *Electrical Eng.* **Index Number:** *7126621*

ME 159-TECHNICAL DRAWING – QUIZ

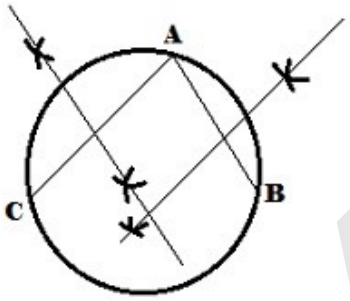
Submission date; Wednesday, 11th May, 2022

ANSWER ALL QUESTIONS

INSTRUCTIONS

Select the correct ANSWER and **circle the letter corresponding to the correct answer in PEN**.

1. Which of the following is not correct in relation to the construction shown in the figure below?



(a) Circumscribing a circle in a triangle.
(b) Constructing a circle to pass through three vertices A, B and C.
(c) Finding the centre of a given circle.
(d) Inscribing a triangle in a given circle.

2. A rod is pivoted at one end and the other end is free to rotate. The locus of the free end after a complete revolution is _____.

(a) an ellipse
(b) a cone
(c) a circle
(d) an involute

3. A locus of a point in a plane for which the sum of its distances from two fixed points in the plane is constant is called _____.

(a) cycloid
(b) ellipse
(c) parabola
(d) trochoid

4. The following are the steps to draw a perpendicular to a line at a point within the line when the point is near the centre of a line. Arrange the steps. Let AB be the line and P be the point in it.

i. P as centre, take convenient radius R_1 and draw arcs on the two sides of P on the line at C, D.
ii. Join E and P.
iii. The line EP is perpendicular to AB.
iv. Then from C, D as centre, take R_2 radius (greater than R_1), draw arcs which cut at E.

(a) i, iv, ii, iii
(b) iii, ii, iv, i
(c) iv, iii, i, ii
(d) ii, i, iv, iii

5. Perpendicular lines cannot be drawn using _____

(a) T- Square
(b) Set-squares
(c) Pro- circle
(d) Protractor

Programme:.....*Electrical Eng.*..... **Index Number:**.....*7126621*.....

ME 159-TECHNICAL DRAWING – QUIZ

Submission date; Wednesday, 11th May, 2022

6. How many pairs of parallel lines are there in regular Hexagon?
   (a) 2 (c) 6
   (b) 3 (d) 1

7. The following are the steps to construct an equilateral triangle when the altitude of a triangle is given. Using, T-square, set-squares only. Arrange the steps.
   i. Join R, Q; T, Q. Q, R, T is the required triangle.
   ii. With a T-square, draw a line AB of any length.
   iii. From a point P on AB draw a perpendicular PQ of given altitude length.
   iv. With 30°–60° set-squares, draw a line making 30° with PQ at Q on both sides cutting at R, T.
   (a) i, iv, ii, iii (c) iv, iii, i, ii
   (b) iii, ii, iv, i (d) ii, iii, iv, i

8. The following are the steps to construct an equilateral triangle, with help of a compass, when the length of altitude is given. Arrange the steps.
   i. Draw a line AB of any length. At any point P on AB, draw a perpendicular PQ equal to altitude length given.
   ii. Draw bisectors of CE and CF to intersect AB at R and T respectively. QRT is required triangle.
   iii. With center Q and any radius, draw an arc intersecting PQ at C.
   iv. With center C and the same radius, draw arcs cutting the 1st arc at E and F.
   (a) i, iii, iv, ii (c) iv, iii, i, ii
   (b) iii, ii, iv, i (d) ii, iii, iv, i

9. To draw an arc of given radius touching two straight lines at right angles with each other, we first draw _____ with centre at the intersection point of the two lines on both the lines.
   (a) An arc (c) A square
   (b) A circle (d) A triangle

10. The following are the steps to draw a tangent to any given circle at any point P on it. Arrange the steps.
   i. Draw the given circle with center O and mark the point P anywhere on the circle.
   ii. With centers O and Q draw arcs with equal radius to cut each other at R.
   iii. Join R and P which is the required tangent.
   iv. Draw a line joining O and P. Extend the line to Q such that OP = PQ.
   (a) i, iv, ii, iii (c) iii, i, iv, ii
   (b) iv, i, iii, ii (d) ii, iv, i, iii

Programme:.....*Electrical Eng.*..... **Index Number:**.....*7126621*.....

ME 159-TECHNICAL DRAWING – QUIZ

Submission date; Wednesday, 11th May, 2022

11. The following are the steps to draw a tangent to given circle from any point outside the circle. Arrange the steps.
    i. With OP as diameter, draw arcs on circle at R and R_1.
    ii. Draw the given circle with center O.
    iii. Join P and R which is one tangent and PR_1 is another tangent.
    iv. Mark the point P outside the circle.
    (a) ii, iv, iii, i
    (b) iv, i, iii, ii
    (c) iii, i, iv, ii
    (d) ii, iv, i, iii

12. How many external tangents are there for two circles?
    (a) 1
    (b) 2
    (c) 3
    (d) 4

13. There are 2 circles say A, and B. A has 20 units radius and B has 10 units radius and distance from centers of A and B is 40 units. Where will be the intersection point of external tangents?
    (a) to the left of two circles
    (b) to the right of the two circles
    (c) middle of the two circles
    (d) they intersect at midpoint of line joining the centers

14. In the general method of drawing an ellipse, a vertical line called ____ is drawn first.
    (a) Tangent
    (b) Normal
    (c) Major axis
    (d) Directrix

15. In arcs of circles method, the foci are constructed by drawing arcs with one of the ends of the ____ axis as center and radius equal to half the length of the ____ axis.
    (a) Minor, major
    (b) Major, major
    (c) Minor, minor
    (d) Major, minor

16. _____ is a curve generated by a point on the circumference of a circle which rolls without slipping on a straight line.
    (a) Trochoid
    (b) Epicycloid
    (c) Cycloid
    (d) Involute

17. The following steps are given to find the normal and tangent for a cycloid. Arrange the steps if C is the centre for generating circle and PA is the directing line. N is the point on cycloid.
    i. Through M, draw a line MO perpendicular to the directing line PA and cutting at O.
    ii. With centre N and radius equal to radius of generating circle, draw an arc cutting locus of C, at M.
    iii. Draw a perpendicular to ON at N which is tangent.
    iv. Draw a line joining O and N which is normal.
    (a) iii, i, iv, ii
    (b) ii, i, iv, iii
    (c) iv, ii, i, iii
    (d) i, iv, iii, ii

Programme:.....*Electrical Eng.*..... **Index Number:**.....*7126621*.....

ME 159-TECHNICAL DRAWING – QUIZ

Submission date; Wednesday, 11th May, 2022

18. The following steps are given to draw tangent and normal to the involute of a circle (center is C) at a point N on it. Arrange the steps.
    i. With CN as diameter describe a semi-circle cutting the circle at M.
    ii. Draw a line joining C and N.
    iii. Draw a line perpendicular to NM and passing through N which is tangent.
    iv. Draw a line through N and M. This line is normal.
    (a) ii, i, iv, iii
    (b) iii, i , iv, ii
    (c) i, iii, iv, ii
    (d) iv, iii, i, ii

19. The sections cut by a plane on a right circular cone are called _____
    (a) Parabolic sections
    (b) Conic sections
    (c) Elliptical sections
    (d) Hyperbolic sections

20. While cutting, if the plane is at an angle and it cuts all the generators, then the conic formed is called a/an _____
    (a) Circle
    (b) Ellipse
    (c) Parabola
    (d) Hyperbola

21. When the projectors are parallel to each other and also perpendicular to the plane, the projection is called _____
    (a) Perspective projection
    (b) Oblique projection
    (c) Isometric projection
    (d) Orthographic projection

22. The object we see in our surrounding usually without drawing came under which projection?
    (a) Perspective projection
    (b) Oblique projection
    (c) Isometric projection
    (d) Orthographic projection

23. In orthographic projection, each projection view represents how many dimensions of an object?
    (a) 1
    (b) 2
    (c) 3
    (d) 0

24. In orthographic projection, an object is represented by two or three views on different planes which _____
    (a) gives views from different angles from different directions
    (b) are mutually perpendicular projection planes
    (c) are parallel along one direction but at different cross-section
    (d) are obtained by taking prints from 2 or 3 sides of object

25. To represent the object on paper by orthographic projection, the horizontal plane (HP) should be placed in which way?
    (a) The HP is turned in a clockwise direction up to 90 degrees
    (b) The HP is turned in anti-clockwise direction up to 90 degrees
    (c) The HP plane is placed to left side of vertical plane parallel to it
    (d) The HP plane is placed to right side of vertical plane parallel to it

Programme: *Electrical Eng.* **Index Number:** *7126621*

ME 159-TECHNICAL DRAWING – QUIZ

Submission date; Wednesday, 11th May, 2022

26. The hidden parts inside or back side of object while represented in orthographic projection are represented by which line?
    (a) Continuous thick line
    (b) Continuous thin line
    (c) Dashed thin line
    (d) Long-break line

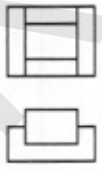
27. The top view of an object is shown on which plane?
    (a) Profile plane
    (b) Vertical plane
    (c) Horizontal plane
    (d) Parallel plane

28. Isometric view of cube is drawn and faces of cube are seen as _____
    (a) square
    (b) rectangle
    (c) rhombus
    (d) parallelogram

29. Drawing pencils are graded according to increase in relative _____
    (a) diameter
    (b) sharpness
    (c) length
    (d) hardness

30. In 3rd angle projection, the _____ lies between _____ and _____
    (a) object, projection plane, observer
    (b) projection plane, object, observer
    (c) reference line, side view, front view
    (d) reference line, left side view, right side view

31. For the multi-view projections shown below, what is the correct corresponding isometric view?



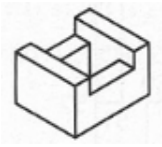
(a)



(b)



(c)



(d)



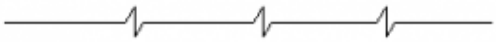
32. Centre lines, section lines are drawn using __ pencil.
    (a) H
    (b) 2H
    (c) 3H
    (d) H

Programme: Electrical Eng. **Index Number:**7126621

ME 159-TECHNICAL DRAWING – QUIZ

Submission date; Wednesday, 11th May, 2022

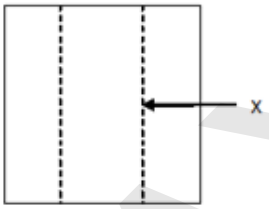
33. The line given below is used for_____



(a) Long-break line
(b) Cutting planes
(c) Centroid lines
(d) Outlines of adjacent parts

34. In engineering drawing, which type of line indicates that there is a change of plane?
(a) Continuous thin wavy
(b) Long chain thin
(c) Continuous thick
(d) Medium thick short dashes

35. From the below figure, what is the name of the line X?



(a) Outline
(b) Section line
(c) Hidden line
(d) Hatching

36. What is the difference between the section line and the centre line?
(a) The length of the long dashes
(b) The length of short dashes
(c) The width of the gap
(d) The two ends of the lines

37. Initial work and construction lines are drawn using __ pencil.
(a) 3H
(b) 4H
(c) H
(d) 2H

38. What do hidden lines in orthographic projections denote?
(a) Holes or slots
(b) Change of plane
(c) Position of cut
(d) Centre of a circle or cylinder

39. Isometric view of cube is drawn; the angle between the edge of cube and vertical will be_____
(a) 15 degrees
(b) 120 degrees
(c) 60 degrees
(d) 30 degrees

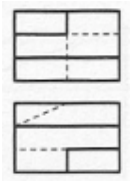
b

Programme:.....*Electrical Eng.*..... **Index Number:**.....*7126621*.....

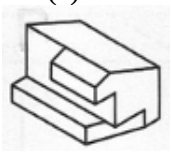
ME 159-TECHNICAL DRAWING – QUIZ

Submission date; Wednesday, 11th May, 2022

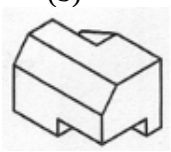
40. For the multi-view projections shown below, what is the correct corresponding isometric view?



(a)



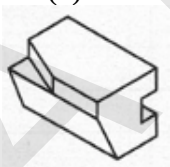
(b)



(c)



(d)



NKAY

ME 159 QUIZ ANSWERS

1	**A**		21	**D**
2	**C**		22	**A**
3	**B**		23	**Bonus b**
4	**A**		24	**B**
5	**C**		25	**A**
6	**B**		26	**A**
7	**D**		27	**C**
8	**A**		28	**C**
9	**A**		29	**D**
10	**A**		30	**B**
11	**D**		31	**B**
12	**B**		32	**C**
13	**B**		33	**A**
14	**D**		34	**C**
15	**A**		35	**C**
16	**C**		36	**D**
17	**B**		37	**C**
18	**A**		38	**A**
19	**B**		39	**C**
20	**B**		40	**B**

NKAY